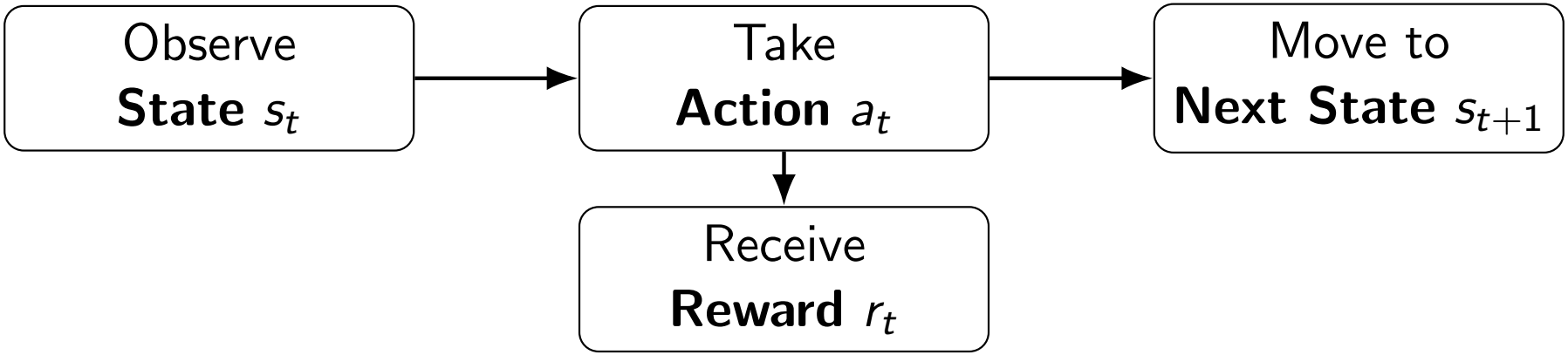
Model-Based Exploration in Markov Decision Processes

Seminar on Reinforcement Learning

Unine

March 3, 2026

MDPs

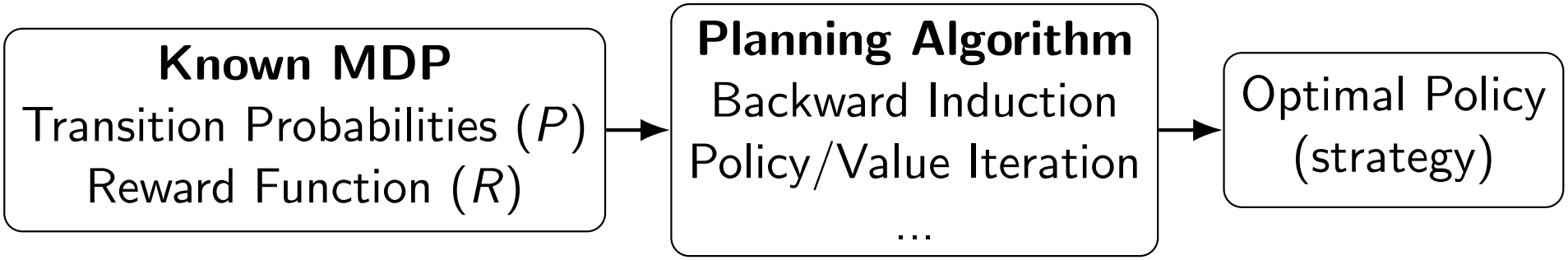


An MDP describes sequential decision making under uncertainty.

Objective: Maximize rewards

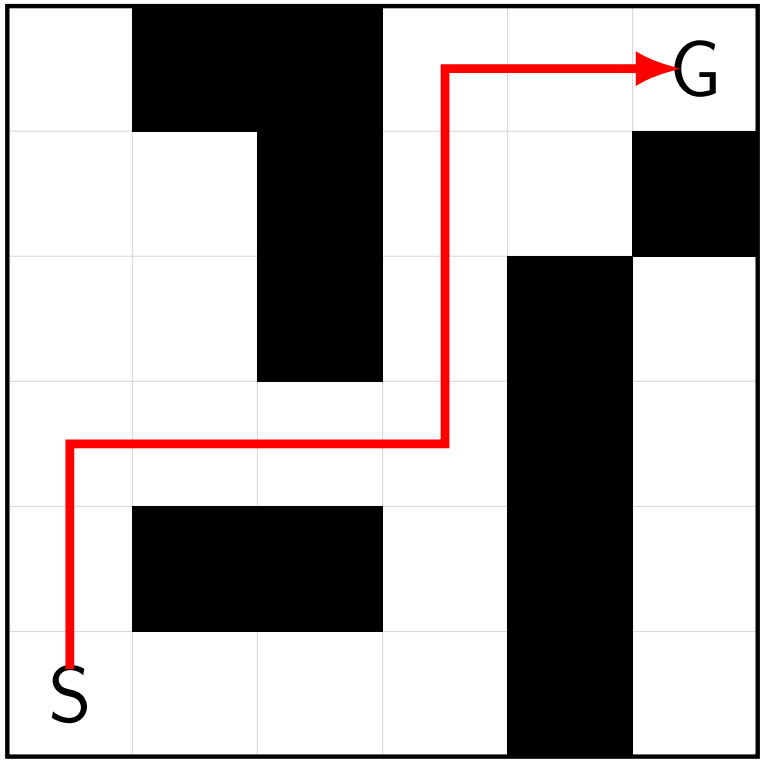
$$\max \sum_t r_t$$

Knowing the MDP = Simple Planning



No exploration. No uncertainty. No data collection.

Knowing the MDP = Solving a Known Maze

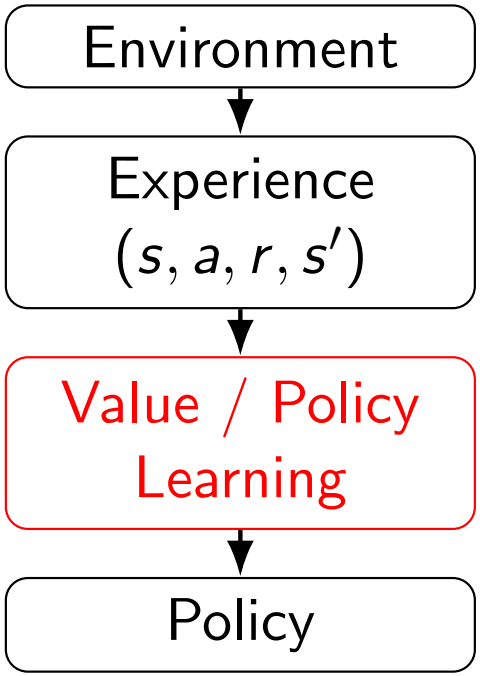


If the layout is known, we simply compute the shortest path.

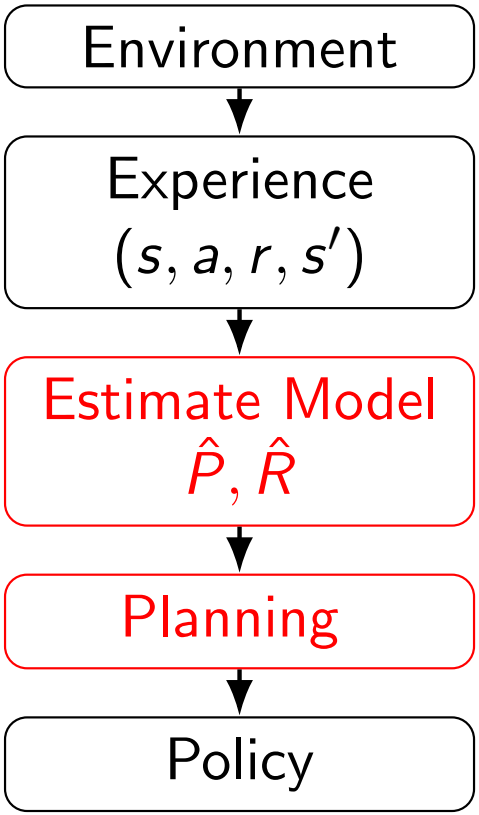
Similarly: If the MDP is known, reinforcement learning reduces to planning.

Unkown MDPs

Model-Free:
Learn directly from data.



Model-Based:
First learn the dynamics, then plan.



The RL Exploration-Exploitation Dilemma

RL

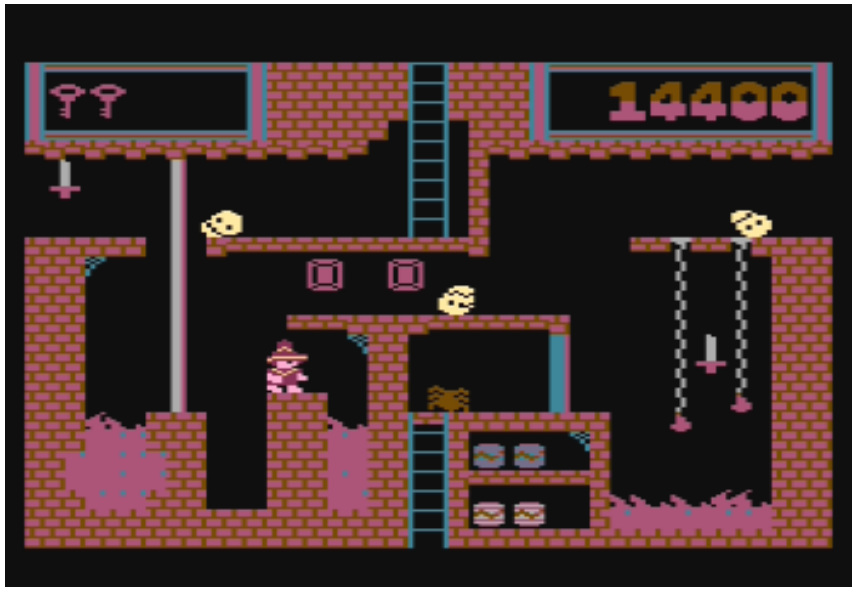
Exploitation
Maximize reward
given current knowledge

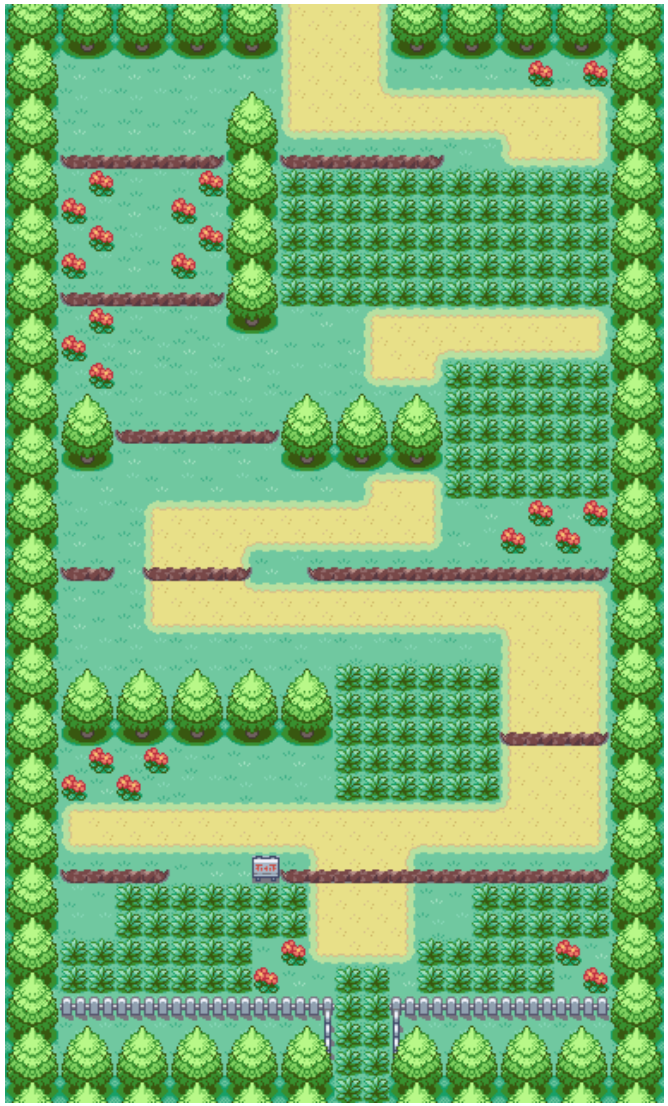
Exploration
Try unknown actions and states,
Improve value / model estimates.

No Exploration:
Some states/actions are never visited
$\rightarrow$ policy may be arbitrarily bad

RL must balance exploration and exploitation to learn an effective policy.

Hard Exploration Games





Motivation: Exploration in RL

Efficient exploration is crucial in unknown MDPs.

Model free exploration:

- Epsilon-greedy Q-learning
  - Sample random actions with small probability.
  - **Very Inefficient Exploration !**
  - **Optimality guarantees are in exponential time !**

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We want **theoretically grounded exploration**.

Model-based exploration:

- Reason about reward/transition uncertainty
- Explore uncertain areas of the MDP systematically.
- **Polynomial time guarantees !**

Historical Context on Model-Based Exploration

- Early ideas with polynomial guarantees (2002):
  - **E**^{3} (Explicit Explore or Exploit), **R-Max**.
  - Pick the least attempted action at underexplored states.
  - Make underexplored state-actions maximally rewarding.

- Model-based Interval Estimation (MBIE, 2008) improves over E^3 and R-Max:
  - Uses **confidence intervals**.
  - Smoothly shrinks optimism as states are visited.

MBIE: Key Idea

1. Maintain empirical estimates for each (s, a):

$$N(s,a), \quad \hat{P}(s'|s,a), \quad \hat{R}(s,a).$$

2. Build confidence intervals around transitions and rewards:

$$|P(s'|s,a) - \hat{P}(s'|s,a)| \leq O\Big(\sqrt{\frac{\log(1/\delta)}{N(s,a)}}\Big)$$

3. Solve the **optimistic MDP** within these intervals:
   - Act as if the world is as rewarding as plausibly possible.

Project content

Phase 1: Implementation

- Reproduce results from the MBIE paper:
  - Implement MBIE and its variant MBIE-EB (exploration bonus).
  - Implement baselines: E^3, R-max
  - Benchmark algorithms on RiverSwim and SixArms MDPs.

Phase 2: Extension options

- Novel exploration schemes.
- Model-free exploration.
- Scaling to larger problems.
- Compare with additional baselines.
- Benchmark on larger/harder MDPs.
- etc.

References

- Strehl, Littman. **An Analysis of Model-Based Interval Estimation for MDPs** (MBIE). Journal of Computer and System Sciences, 2008.
- Brafman, Tennenholtz. **R-Max: A General Polynomial Time Algorithm for Near-Optimal Reinforcement Learning** (R-max). JMLR, 2002.
- Kearns, Singh. **Near-Optimal Reinforcement Learning in Polynomial Time** (E^3). Machine learning, 2002.